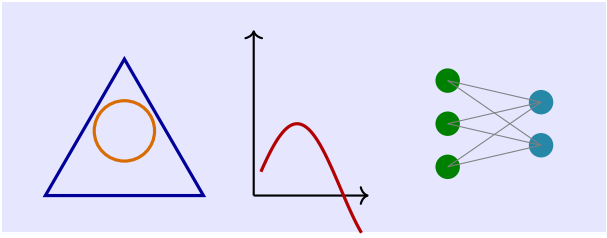
TikZ Gallery

Vector diagrams for mathematics, physics,
chemistry and computer science



N'solé Samson Yamontche
https://samsonyamontchensolegit.github.io

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Every figure in this document is drawn directly in LaTeX with TikZ and its libraries: no external images, no raster graphics. The drawings scale without loss of quality and use the same fonts as the text.

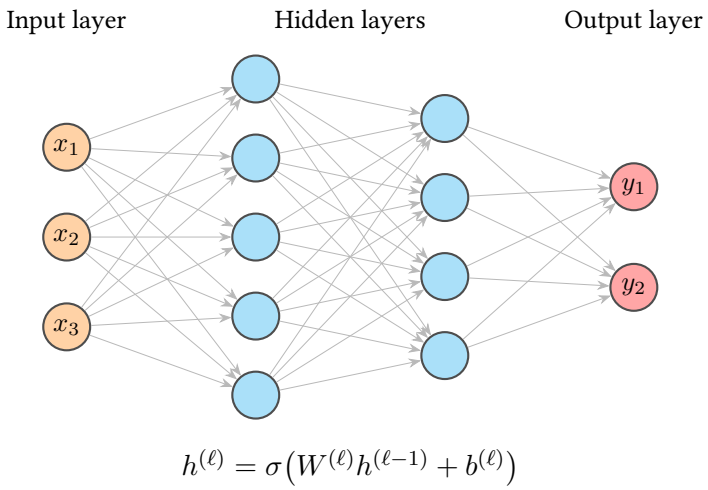


Figure 1: A feed-forward neural network with two hidden layers. *Libraries:* arrows.meta.

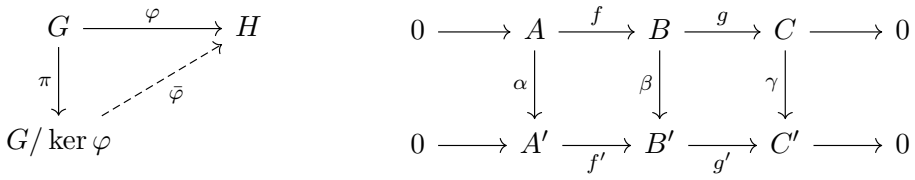


Figure 2: Commutative diagrams: the first isomorphism theorem (left) and a morphism of short exact sequences (right). *Libraries:* tikz-cd.

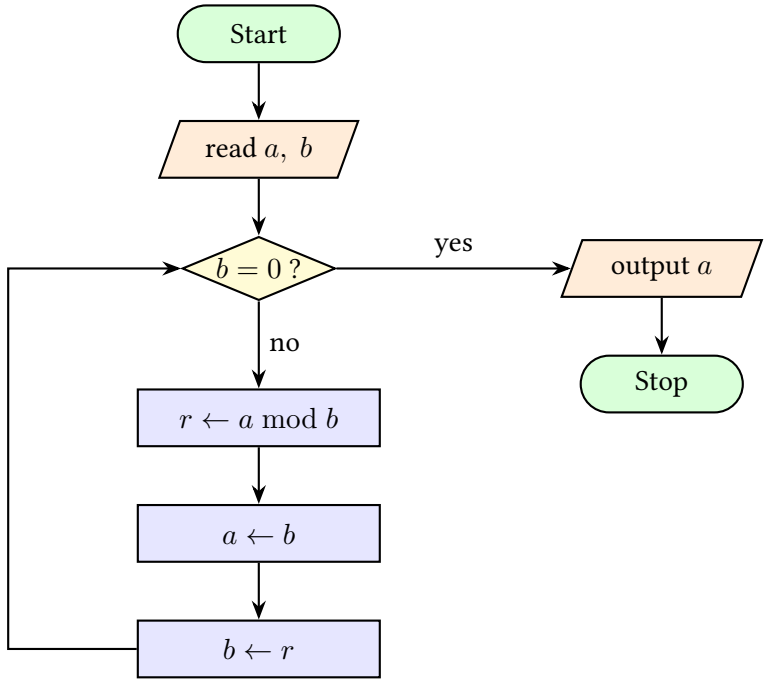
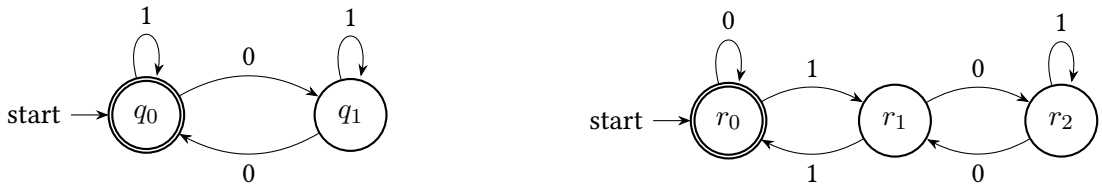


Figure 3: Flowchart of Euclid's algorithm for the greatest common divisor. *Libraries:* shapes.geometric, positioning.



(a) Even number of 0s.

(b) Binary numbers divisible by 3.

Figure 4: Two deterministic finite automata. *Libraries:* automata.

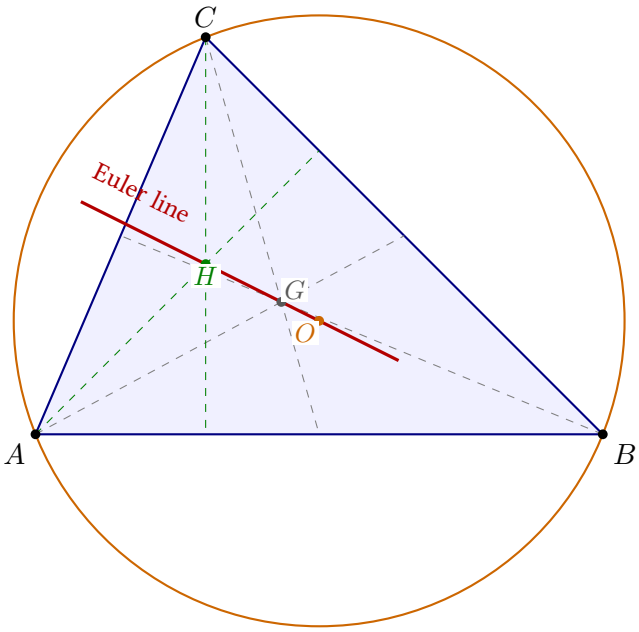


Figure 5: A triangle with its circumcircle, medians and altitudes. The circumcentre O, the centroid G and the orthocentre H lie on the Euler line, with $OG : GH = 1 : 2$. *Libraries:* calc.

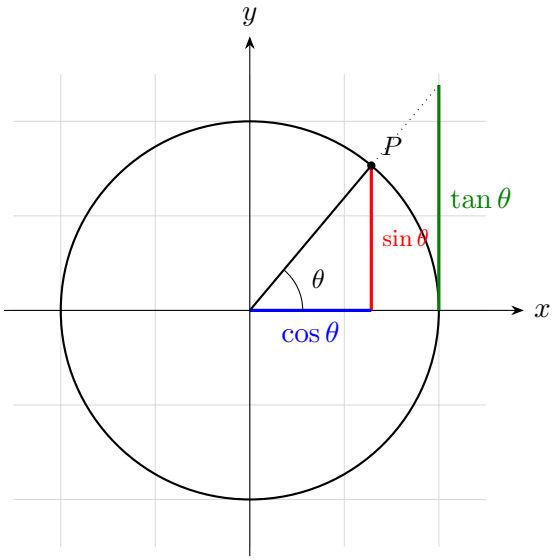
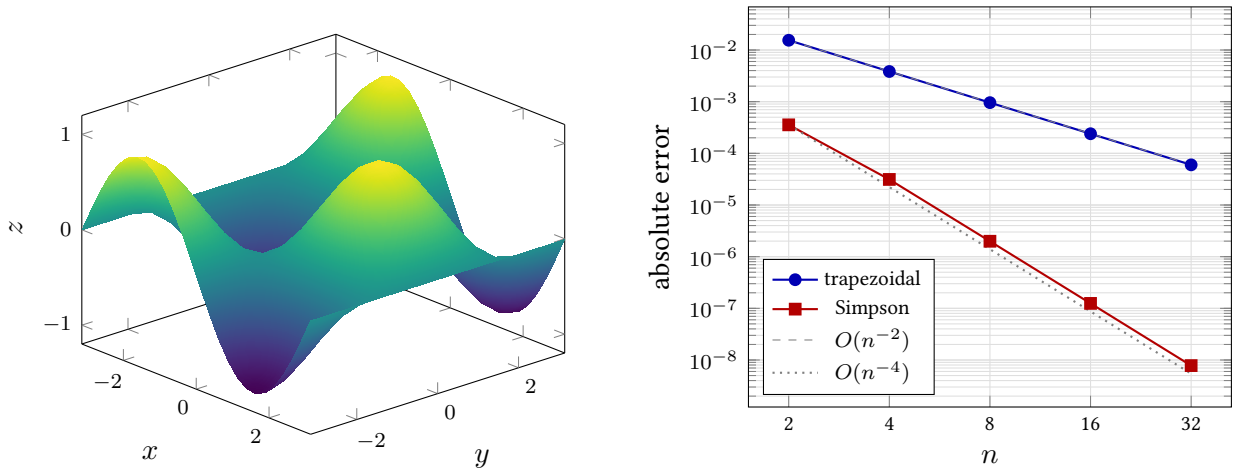


Figure 6: The unit circle with the three basic trigonometric functions of the angle $\theta = 50°$, with $P = (\cos\theta, \sin\theta)$. *Libraries:* standard TikZ.



(a) The surface $z = \sin x \cos y$.

(b) Errors for $\int_0^1 e^{-x^2}\,\mathrm{d}x$.

Figure 7: Plots with `pgfplots`: a 3D surface (left) and a log–log convergence study with the reference slopes (right). *Libraries:* pgfplots.

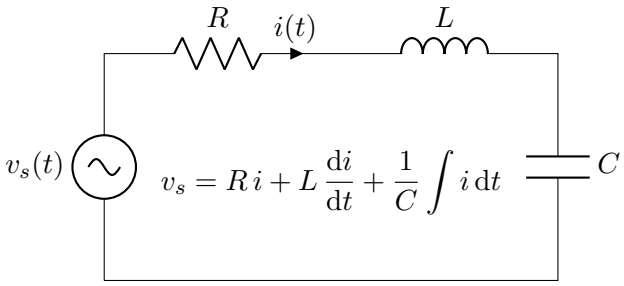


Figure 8: A series RLC circuit driven by a sinusoidal voltage source. *Libraries:* circuitikz.

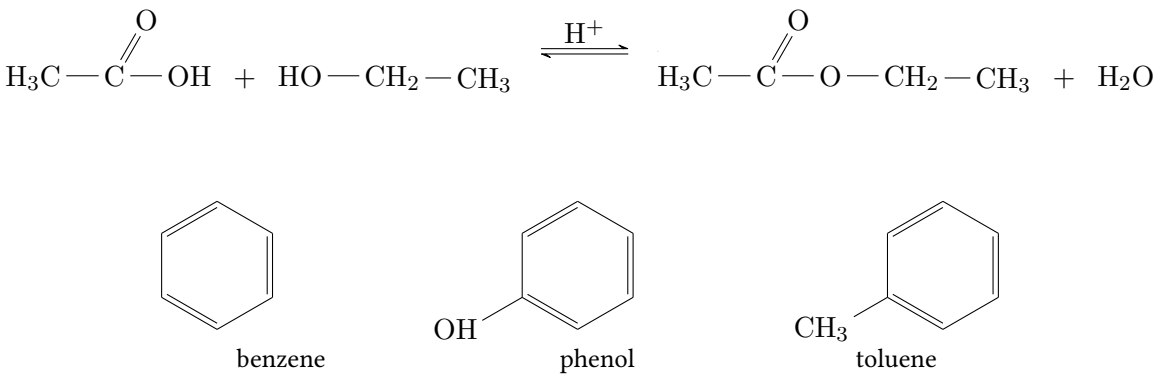


Figure 9: Fischer esterification of acetic acid with ethanol, and three aromatic compounds. *Libraries:* chemfig.

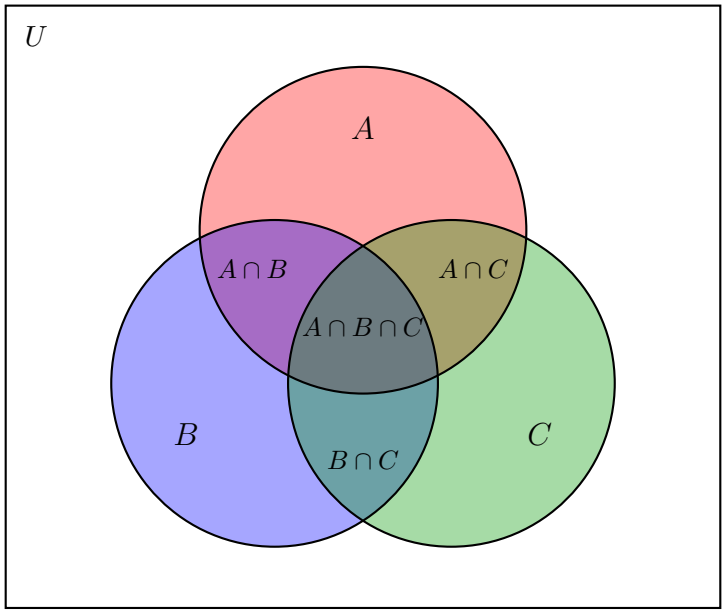


Figure 10: A Venn diagram of three sets inside a universe U. *Libraries:* standard TikZ.

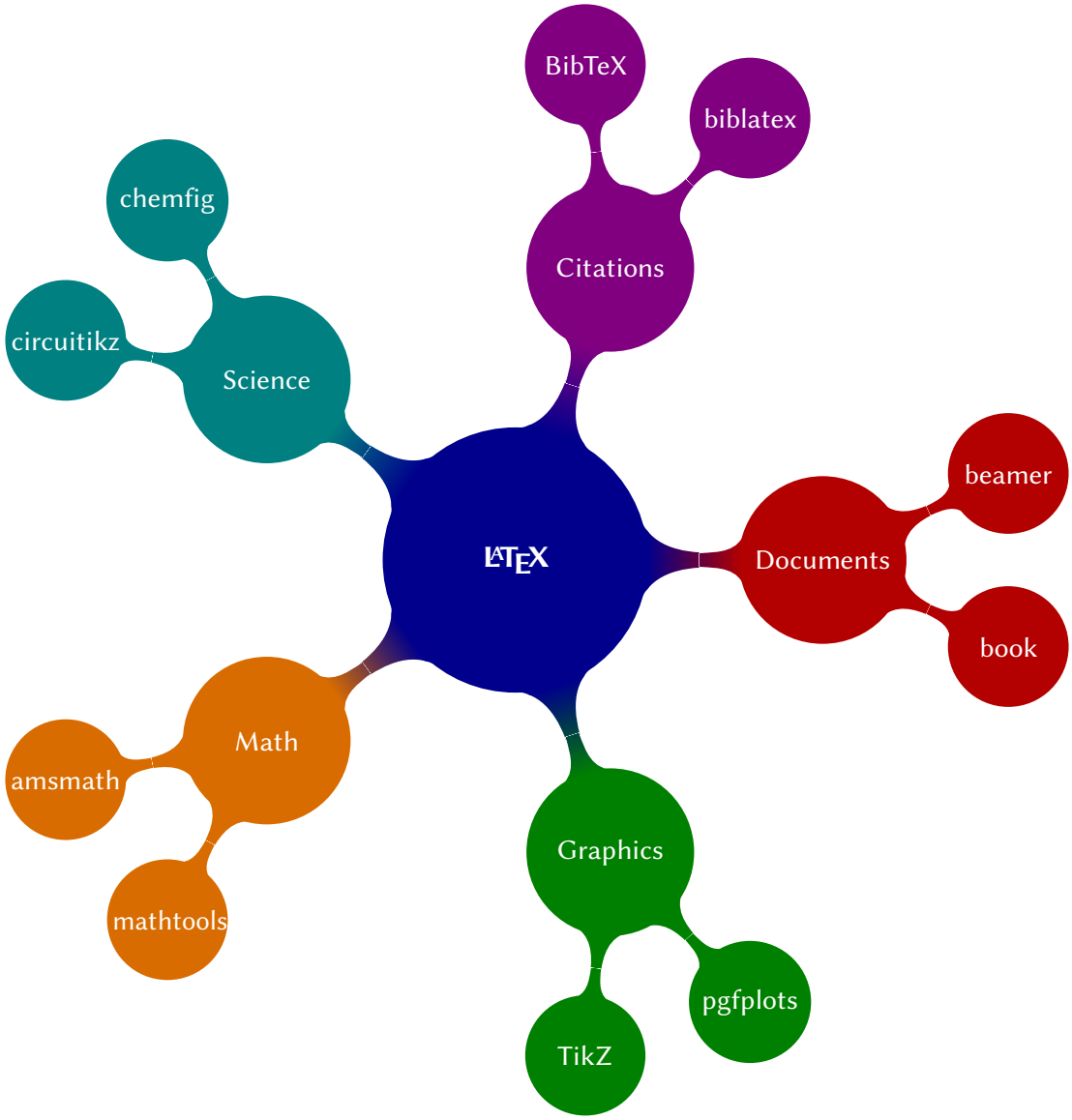


Figure 11: A mind map of the LaTeX ecosystem. *Libraries:* mindmap, trees.

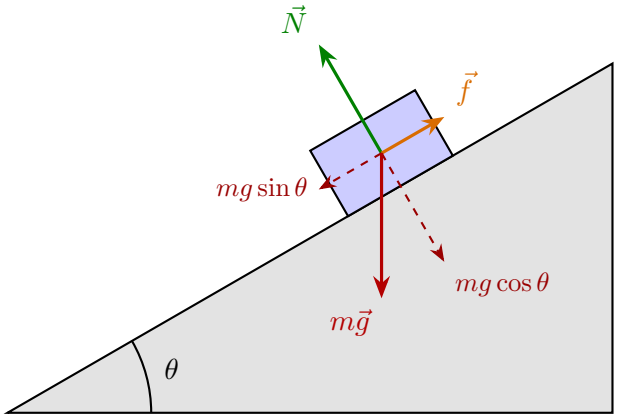


Figure 12: Free-body diagram of a block at rest on an inclined plane: $N = mg\cos\theta$ and $f = mg\sin\theta$. *Libraries:* standard TikZ.